

The flash that wouldn't stay

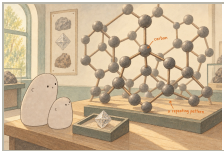
The flash that wouldn't stay. • Curiosity day

Name _____

Date _____



A flash winks from a museum case. "The diamond makes light!" says Little Wonderer. Big points to its deep beginning. Heat and huge pressure helped carbon grow into a crystal far underground—without lamp or sun. "Then where does the flash come from?"



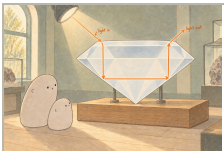
"Maybe light hides in the tiny parts," Little says. Balls and rods stand for carbon atoms and their connections. The pattern repeats through a diamond—but it is no lantern. "We found the stone," says Big, "not the flash."



Little traces an arrow up. "To meet light, it had to leave the dark!" Long after some diamonds formed, volcanic eruptions carried them toward the surface. "Now it can meet light," says Little. Big nods. "But meeting light is not making it."



A label marked 10 stops Little. Diamond tops the Mohs scratch scale. "So nothing can hurt it?" Big explains that a diamond can still chip or break. "Hard to scratch is not unbreakable," Little says, "and it still doesn't explain the flash."



At the last model, a lamp clicks on. "There!" Little cries. Light enters the diamond, bends, bounces, and comes out again. Cut, flat faces called facets guide its path. Little follows the line. "The diamond sends borrowed light back!"



Little shifts. One bright face dims; another flashes. "Found you!" The flash was not stored in the stone. It happens when light, cut facets, and watching eyes meet along the right paths. Little sketches both spots, leaving room for a new question.

Think about it: What did Little first think made the flash? Which clue changed that idea? At the end, what had to meet for Little to see a flash?